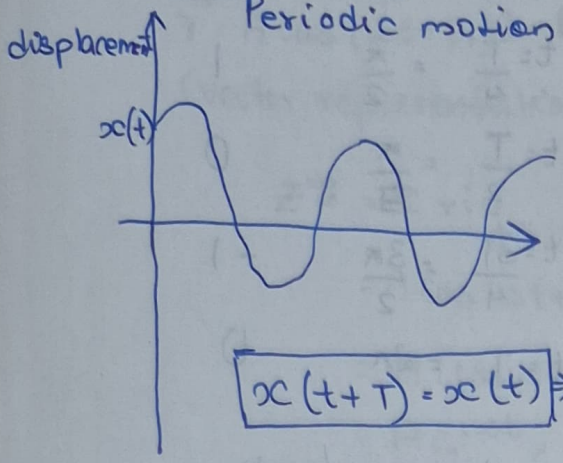
WAVES

Oscillation: Bounded periodic motion.
Periodic motion in time.

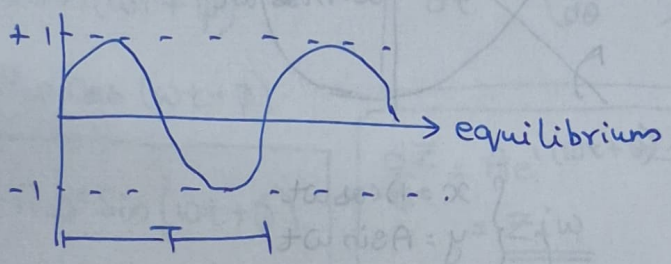


$x(t+T) = x(t)$ ⇒ periodic osc., or vibration.

Simple harmonic oscillation / Motion
SHO/SHM

$x(t) = A\sin\omega t$ or $A\cos\omega t$

$x(t) = A\sin\omega t$



$\max(\sin\omega t) = +1 \Rightarrow \max(x(t)) = A$
$\min(\sin\omega t) = -1 \Rightarrow \min(x(t)) = -A$

$-A \le x \le A$
$x \in [-A, A]$

Connection b/w ω & T

* Comes from periodicity condition.

$x(t+T) = xt$
$A\sin\omega(t+T) = A\sin\omega t$
$\sin(\omega t + \omega T) = \sin\omega t$
$\sin(\theta + \alpha) = \sin\theta$
$\alpha = 0, \pm 2\pi, \pm 4\pi, \ldots$

$\alpha = \omega T = 2\pi$

$$T = \frac{2\pi}{\omega}$$

T = period (sec)

$\frac{1}{T}$ = number of cycles per sec. = frequency = ν

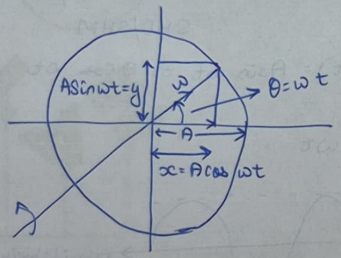
$$\nu = \frac{1}{T} = \frac{\omega}{2\pi}$$

$$\omega = 2\pi\nu = \frac{2\pi}{T}$$

$$\omega t = \frac{2\pi t}{T}$$

	$\omega t = 0$	$\sin\theta = 0$
$t = 0$	$= 0$	$\omega t = 0$
$t = \frac{T}{4}$	$= \frac{\pi}{2}$	1
$t = \frac{T}{2}$	$= \pi$	0
$t = \frac{3T}{4}$	$= \frac{3\pi}{2}$	−1
$t = T$	$= 2\pi$	0

WAVE: oscillation propogating in space/medium.



$$\begin{cases} x = A\cos\omega t \\ y = A\sin\omega t \end{cases}$$

$\Rightarrow x = A,\ y = 0$ at $t = 0$

$t = 0,\quad \theta(0) = \theta_0 \neq 0$

$\Delta\theta = \theta - \theta_0 = \omega t$

$$\frac{d\theta}{dt} = \omega$$

$$\int_{\theta_0}^{\theta} d\theta = \int_0^t \omega\, dt$$

$$\theta - \theta_0 = \omega t$$

$$\boxed{\theta(t) = \theta_0 + \omega t}$$

$0 \leq \theta < 2\pi$ ~~$\theta \in [0, 2\pi)$~~ $\theta \in [0, 2\pi)$

$$\boxed{x = A\cos(\omega t + \theta_0)}$$

$$\boxed{y = A\sin(\omega t + \theta_0)}$$

$$\vec{r} = A\cos(\omega t + \theta_0)\hat{i} + A\sin(\omega t + \theta_0)\hat{j}$$

(Vector representation)

$$z = x + iy$$

$$= A\cos(\omega t + \theta_0) + iA\sin(\omega t + \theta_0)$$

$$= A(\cos\omega t + i\sin\omega t)$$

$$= Ae^{i\omega t},\ Ae^{i\theta(t)}$$ [Complex or Phasor representation]

$$\boxed{e^{i\theta} = A\cos\theta + iA\sin\theta}$$

$$\theta = \theta_0 + \omega t$$

$$z = x + iy = Ae^{i(\omega t + \theta)}$$

$$x = A\sin(\omega t + \phi)$$

$$\frac{dx}{dt} = A\omega\cos(\omega t + \phi)$$

$$\boxed{\frac{dx}{dt} = A\left(\frac{d}{d\theta}\sin\theta\right)\left(\frac{d(\omega t + \phi)}{dt}\right)}$$

$$\boxed{\frac{dx^2}{dt^2} = -A\omega^2\sin(\omega t + \phi)}$$

$$= -\omega^2 x(t)$$

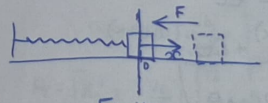
$$\boxed{\frac{dz}{dt} = Ae^{i(\omega t + \theta)} \times i\omega = zi\omega}$$

$$\boxed{\frac{dz^2}{dt^2} = i\omega(i\omega z) = i^2\omega^2 z = -\omega^2 z}$$

Acceleration

$$a \propto -x$$

Ideal Spring



$F = -Kx$

$$m\frac{dx^2}{dt^2} = -Kx$$

$$\frac{dx^2}{dt^2} = -\frac{K}{m}x$$

$$\frac{K}{m} = \omega^2$$

$$\omega = \sqrt{\frac{K}{m}}$$

$\omega^2 =$ Restoring force per unit displacement / m

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

$$X(t) = Ce^{\lambda t},$$

$$\frac{dx^2}{dt^2} = \lambda^2 Ce^{\lambda t}$$

$$(\lambda^2 + \omega^2) \underbrace{Ce^{\lambda t}}_{\text{↳ can't be zero.}} = 0$$

$$\Rightarrow \lambda^2 = -\omega^2$$

$$\lambda_{1,2} = \pm i\omega$$

$$x_1(t) = C_1 e^{i\omega t}, \quad x_2(t) = C_2 e^{-i\omega t}$$

$$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

Note: $e^{\pm i\omega t} = \cos(\omega t) + i\sin(\omega t)$

$$x(t) = A\cos\omega t + B\sin\omega t$$

where A & B are real.

$$x = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$$

$$= A'\sin\omega t + B'\cos\omega t$$

$$B' = C_1 + C_2$$

$$A' = i(C_1 + C_2)$$

$$C_1 = \frac{1}{2}(B' - iA')$$

$$C_2 = \frac{1}{2}(B' + iA') = C_1^*$$

$$x = C_1 e^{i\omega t} + C_1^* e^{-i\omega t}$$

$$C_1 = |C_1| e^{i\alpha} = Ce^{i\alpha}$$

$$x = Ce^{i(\omega t + \alpha)} + Ce^{-i(\omega t + \alpha)}$$

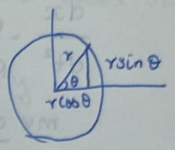
$$e^{i\theta} + e^{-i\theta} = 2\cos\theta$$

$$\theta = \omega t + \alpha$$

$$x = 2C\cos(\omega t + \alpha)$$

$$A_{new} = 2C$$

$$X = A_{new}\cos(\omega t + \alpha)$$



$$r = \sqrt{x^2 + y^2}$$

$$Z = x + iy$$

$$= r\cos\theta + ir\sin\theta$$

$$Z = r(\cos\theta + i\sin\theta)$$

$$Z = re^{i\theta}$$

$$|Z| = \sqrt{r^2(\cos\theta + i\sin\theta)^2}$$

~~$r(\cos\theta + i\sin\theta)$~~

~~$|Z| = r(\cos\theta + i\sin\theta)$~~

$$|Z| = r$$

$$Z = |Z|e^{i\theta}$$

$$x(t)\Big|_{t=0} = X(0) = X_0$$

$$\frac{dx(t))}{dt}\Big|_{t=0} = \dot{X}(0) \equiv X_0 \equiv V_0 \equiv V(0)$$

$$x(t) = A\cos(\omega t + \alpha)$$

$$x(0) = A\cos\alpha = X_0$$

$$\frac{dx(0)}{dt} = \dot{X}(0) = V_0 = -\omega A\sin\alpha$$

$$A\sin\alpha = -\frac{V_0}{\omega} \qquad A\cos\alpha = X_0$$

$$\tan\alpha = \frac{-V_0}{\omega X_0}$$

$$A = \left[X_0^2 + \left(\frac{V_0}{\omega}\right)^2\right]^{1/2}$$

example: $x_0 = 1, v_0 = -1, \omega = 1$

$$A = \sqrt{1^2 + \left(\frac{-1}{1}\right)^2} = \sqrt{2}$$

$$\tan\alpha = 1 \quad \alpha = \frac{\pi}{4}$$

$$x(t) = \sqrt{2}\cos\left(t + \frac{\pi}{4}\right)$$

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = -\omega^2 x$$

$$mv\frac{dv}{dt} = -Kx\cdot\frac{dx}{dt}$$

$$\frac{d}{dt}\left(\frac{mv^2}{2}\right) = -\frac{d}{dt}\left(\frac{K}{2}x^2\right)$$

$$\int_{V_0}^{V}\frac{d}{dt}\left(\frac{mv^2}{2}\right) = -\int_{X_0}^{X}\frac{d}{dt}\left(\frac{K}{2}x^2\right)$$

$$\frac{m}{2}\left(V^2 - V_0^2\right) = \frac{-K}{2}\left(x^2 - x_0^2\right)$$

$$\frac{mv^2}{2} - \frac{mv_0^2}{2} = \frac{-Kx^2}{2} + \frac{Kx_0^2}{2}$$

$$\frac{mv^2}{2} + \frac{Kx^2}{2} = \frac{Kx_0^2}{2} + \frac{mv_0^2}{2}$$

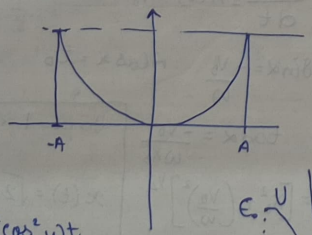
$U = \frac{1}{2}Kx^2$

E_0 = total mechanical energy

$$E = \frac{1}{2}mv_0^2 + \frac{1}{2}Kx^2$$

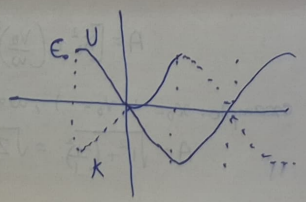
$$E_0 = \frac{1}{2}KA^2 = \frac{1}{2}m\omega^2A^2$$

$$\boxed{E_0 = \frac{1}{2}m(\omega A)^2}$$



$$U = \frac{1}{2}KA^2\cos^2\omega t$$

$$K = \frac{1}{2}m\omega^2\sin^2\omega t$$



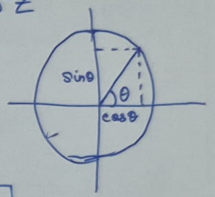
Phasor method of solving SHM from 2nd order ODE

$$\frac{d^2x}{dt^2} = -\omega^2x, \quad \frac{d^2y}{dt^2} = -\omega^2y$$

$$\frac{d^2Z}{dt^2} = -\omega^2Z$$

$$\tilde{V} = \frac{dx}{dt} + i\frac{dy}{dt}$$

$$\frac{dv}{dt} = \frac{d^2x}{dt^2} + i\frac{d^2y}{dt^2} = -\omega^2Z$$

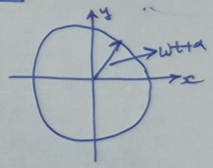


$$Z = Ae^{i(\omega t+\alpha)}$$

$$\frac{dZ}{dt} = i\omega Ae^{i(\omega t+\alpha)}$$

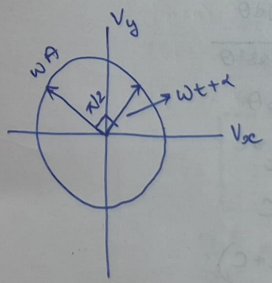
$$\boxed{e^{i\pi/2} = \cos\frac{\pi}{2}^{0} + i\sin\frac{\pi}{2}^{1} = i}$$

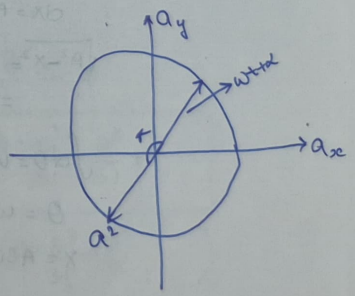
$$\frac{d^2Z}{dt^2} = -\omega^2Ae^{i(\omega t+\alpha+\frac{\pi}{2})}$$



$$-1 = e^{i\pi}$$

$$\frac{d^2Z}{dt^2} = \omega^2Ae^{i(\omega t+\alpha+\pi)}$$

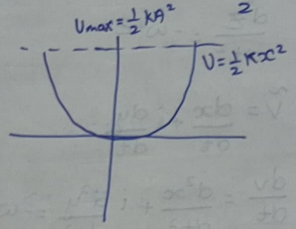




Energy Method

$\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \text{const} = E_0 = \cancel{\text{total}}$ initial energy

$= \frac{1}{2}kA^2$



$v^2 + \frac{k}{m}x^2 = \frac{k}{m}A^2$

$\frac{k}{m} = \omega^2$

$v = \left(\frac{dx}{dt}\right)^2$

$\left(\frac{dx}{dt}\right)^2 = \omega^2(A^2 - x^2)$

$\frac{dx}{dt} = \pm\omega\sqrt{A^2 - x^2}$

$\int \frac{dx}{\sqrt{A^2 - x^2}} = \omega \int dt$

let $x = A\sin\theta$

$dx = A\cos\theta\, d\theta$

$\sqrt{A^2 - x^2} = \sqrt{A^2\cos^2\theta}$

$= A\cos\theta$

$\int d\theta = \omega \int dt$

$\theta = \omega t + C$

$x = A\sin(\omega t + C)$

At extreme left end:

$t = 0, x(0) = -A$

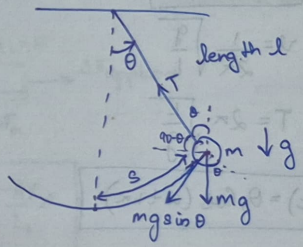
$-A = A\sin C$

$\sin C = -1$

$C = -\frac{\pi}{2}$

$x = A\sin\left(\omega t - \frac{\pi}{2}\right) = \underline{-A\cos\omega t}$

Simple Pendulum



$s = l\theta$

$a = \frac{d^2s}{dt^2}$

$\cancel{m}\frac{d^2s}{dt^2} = -\cancel{m}g\sin\theta$ (-ve coz of direction)

$L\frac{d^2\theta}{dt^2} = -g\sin\theta$

$\frac{d^2\theta}{dt^2} = \frac{-g}{L}\sin\theta$

$$\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \cdots$$

$\frac{d^2\theta}{dt^2} = \frac{-g}{\cancel{g}L}\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \cdots\right)$

Non linear

for $\theta \ll \frac{\pi}{2}$

$\sin\theta \approx \theta$

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$$

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

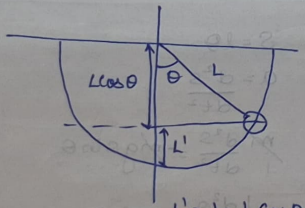
$$\frac{g}{L} = \omega^2$$

$$\boxed{\omega = \sqrt{\frac{g}{L}}}$$

$$\nu = \frac{1}{2\pi}\sqrt{\frac{g}{L}}$$

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$\boxed{\theta(t) = \theta_0 \cos(\omega t + \alpha)}$$



$$L' = L - L\cos\theta$$

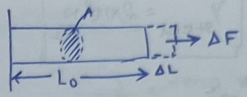
$$U(\theta) = mg(L - L\cos\theta)$$

$$\boxed{U(\theta) = mgL(1-\cos\theta)}$$

$$\cos\theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \cdots$$

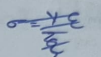
$$1 - \cos(\theta) = \frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} + \cdots$$

$$\approx \frac{1}{2}\theta^2$$



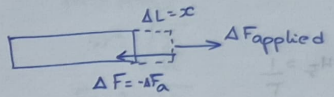
Strain, $\varepsilon = \frac{\Delta L}{L_0}$

Stress, $\sigma = \frac{\Delta F}{A}$



$$\boxed{\sigma = Y\varepsilon}$$

$\frac{\sigma}{\varepsilon} = Y$ = Young's Modulus
= Material Constant



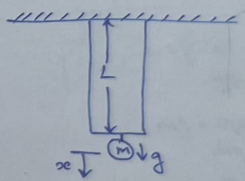
$$\boxed{\Delta F = -\frac{AY}{L_0}x}$$

$$\frac{\Delta F_a}{A\varepsilon} = Y$$

$$\Delta F_a = AY \cdot \frac{\Delta L}{L_0}$$

$$= \frac{AY}{L_0}\Delta L$$

$$= \frac{AY}{L_0}x$$



$$m\frac{dx^2}{dt^2} = -\frac{YA}{L_0}x$$

$$\frac{dx^2}{dt^2} = -\frac{YA}{mL_0}x$$

$$= -\omega^2 X$$

$$\omega = \sqrt{\frac{AY}{mL_0}}$$

$$\nu = \frac{1}{2\pi}\sqrt{\frac{AY}{mL_0}} \quad \& \quad T = 2\pi\sqrt{\frac{AY}{mL_0}}$$

eg: Steel wire = $L_0 = 1m$

$d = 1mm$

$Y_{steel} = 2\times10^{11}\ \frac{N}{m^2}$

$A = \frac{\pi d^2}{4} \approx 0.8\times10^{-6}\ m^2$

$K = \frac{AY}{L_0} = \frac{0.8\times10^{-6}\times2\times10^{11}}{1}$

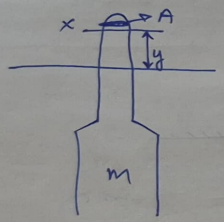
$= 1.65\times10^{5}\ N/m$

$m = 1kg$

$\omega^2 = \frac{K}{m} \approx 16\times10^4\ s^{-2}$

$\omega \approx 400\ s^{-1}$

$\nu = \frac{\omega}{2\pi} \approx 60Hz = \frac{1}{T}$



In eqm

$m\frac{d^2y}{dt^2} = -mg + g\rho V^\circ_{liq}$

$= 0$

only valid for $y = 0$

V°_{liq} = Volume of liquid displaced in eqm

$mg = g\rho V^\circ_l$

$y \neq 0$

$V_{liq} = V^\circ_{liq} - Ay$

$V^\circ_{liq} = V_{liq} + Ay$

$m_l = \rho V_l$

$= \rho(V^\circ_{liq} - Ay)$

~~$m\frac{d^2y}{dt^2} = -mg + g\rho(V_{liq} + Ay)$~~

for any position

$m\frac{d^2y}{dt^2} = -mg + g\rho V_{liq}$

$\frac{d^2y}{dt^2} = -\frac{g\rho A}{m}y$

$\omega = \sqrt{\frac{g\rho A}{m}}$

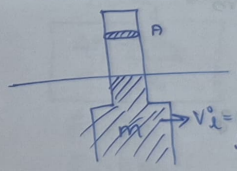
Problem:

Ships =

$m \approx Ah\rho$

$\omega = \sqrt{\frac{g}{h}}$

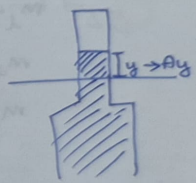
$T = 2\pi\sqrt{\frac{h}{g}}$



V°_l = eqm Vol of liq displaced

m°_l = eqm mass of liq displaced

$= \rho V^\circ_l$



$V_l = V^\circ_l - Ay$

$m_l = \rho V_l$

$= \rho(V^\circ_l - Ay)$

$m\frac{d^2y}{dt^2} = -mg + m_l g$

$= (m_l - m)g$

Ineqm

$y=0, V=0$

$\frac{d^2y}{dt^2}=0 \Rightarrow F_g=0$

$y=0, m_l=m_l^\circ$

$0=(m_l^\circ-m_l)g$

$m_l^\circ=m$

$\rho V_l^\circ=m$

$V_l^\circ=\frac{m}{\rho}$

$m_l^* = \rho V_l^\circ - \rho A y$

$= m-\rho A y$

$m_l - m = m-\rho A y - m$

$m_l - m = -\rho A y$

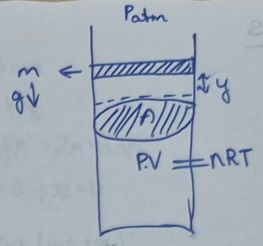
$m\frac{d^2y}{dt^2} = (m_l-m)g$

$= -\rho g A y$

$= -Ky$

$K=\rho g A$

$\omega^2=\frac{K}{m}=\frac{\rho g A}{m}$



$P=P_{atm}+\frac{mg}{A}$

let $T=$ cst

$PV=$ cst

diff (UV)

$P\Delta V + V\Delta P = 0$

$P\Delta V = -V\Delta P$

$\Delta P = -\frac{P\Delta V}{V}$

$P=-V\frac{\Delta P}{\Delta V}$,

Isothermal Bulk Modulus

$P=\frac{-\Delta P}{\left(\frac{\Delta V}{V}\right)}$

$\boxed{P=-V\left(\frac{\partial P}{\partial V}\right)_T = B_T}$

★ Continued ahead →

$\Delta V = Ay$ (when piston displaced by y)

$\boxed{\Delta P = -\frac{P_0 A y}{V_0}}$

$F=\Delta P A$

$F=-\frac{P_0 A y}{L_0}$

$m\frac{d^2y}{dt^2}=-\frac{P_0 A y}{L_0}$

$\frac{K}{m}=\frac{P_0 A}{mL_0}$

Problems

1-10 $y = A\cos(kx) + B\sin(kx)$

$$\frac{dy}{dx} = -Ak\sin kx + Bk\cos kx$$

$$\frac{d^2y}{dx^2} = -Ak^2\cos kx - Bk^2\sin kx$$

also $\frac{d^2y}{dx^2} = -ky$

$$-ky = -k(A\cos kx + B\sin kx)$$

$$y = A\cos kx + B\sin kx$$

$$y = k(A\cos kx + B\sin kx)$$

$$y = ky$$

1-11) $A = 5\text{ cm}$

$f = 1\text{ Hz}$

$\omega = 2f\pi = 2\pi\text{ rad/s}$

at $t = 0$; $x = 0$

$x = A\cos(\omega t + \alpha)$

a) $0 = 5\cos\alpha$

α can be $\frac{\pi}{2}$ or $\frac{3\pi}{2}$

$x = 5\cos\left(2\pi t + \frac{\pi}{2}\right)$ or $x = 5\cos\left(2\pi t + \frac{3\pi}{2}\right)$

$$P = -V\left(\frac{\partial P}{\partial V}\right)_T = B_T$$

$$P = \frac{nRT}{V}$$

$$\frac{\partial P}{\partial V} = \frac{-nRT}{V^2}$$

$$\boxed{-V\frac{\partial P}{\partial V} = \frac{nRT}{V} = P}$$

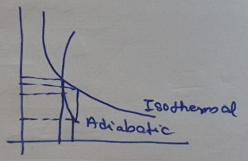
When temp is constant the bulk modulus of an Ideal gas is just Preassure.

B_S = Bulk modulus (entropy constat) ⇒ Adiabatic

$$B_S = -V\left(\frac{\partial P}{\partial V}\right)_S$$

$$PV^\gamma = \text{constant} = C$$

$$P = \frac{C}{V^\gamma} = CV^{-\gamma}$$



$$\left(\frac{\partial P}{\partial V}\right)_S = -\gamma C V^{-\gamma-1}$$

$$\left(\frac{\partial P}{\partial V}\right)_S = \frac{-\gamma C V^{-\gamma}}{V}$$

$$V\left(\frac{\partial P}{\partial V}\right)_S = -\gamma P$$

$$\boxed{-V\left(\frac{\partial P}{\partial V}\right)_S = \gamma P}$$

$$B_S = -V\left(\frac{\partial P}{\partial V}\right)_S$$

$$= -V\frac{\Delta P}{\Delta V}$$

$$\Delta P = -B_S\left(\frac{\Delta V}{V}\right)$$

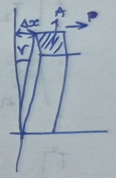
$$F = \Delta P A = \frac{-AB_S A_y}{V_0 \to AL_0}$$

$$F = \frac{-B_S A y}{L_0} = -m\frac{d^2y}{dt^2}$$

$$\frac{d^2y}{dt^2} = \frac{-B_S A}{mL_0}$$

$$\omega = \sqrt{\frac{\gamma P_0 A}{mL_0}}$$

Torsional Oscillation



$$\tau_{shear} = \frac{\Delta F}{A}$$

γ = shear stress = $\frac{\Delta x}{L}$

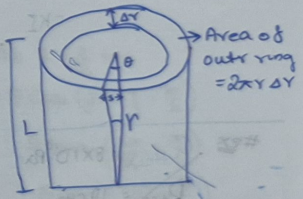
$$\frac{\tau_{shear}}{\gamma} = G \text{ (rigidity modulus)}$$

$$\frac{\tau_{shear}}{r} = \frac{\Delta F}{A r}$$

$$G = \frac{\Delta F}{A r}$$

$$GA\gamma = \Delta F$$

$$\Delta F = G\gamma\Delta A$$



$$\gamma = \frac{s}{L} = \frac{r\theta}{L}$$

$$\Delta F = -G\frac{r\theta}{L}\cdot 2\pi r\Delta r$$

Restoring →

Torque $\Delta\tau = r\Delta F = G\frac{\theta}{L}\cdot 2\pi r^3 dr$

$$\tau = \frac{G\theta}{L}\cdot 2\pi\int_0^R r^3 dr$$

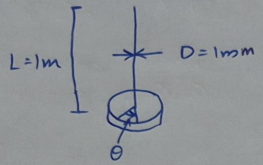
$$\tau = -\frac{\pi G R^4}{2L}\theta$$

$$\tau = -C\theta$$

$$\tau = I\frac{d^2\theta}{dt^2} = -C\theta$$

$$\omega = \sqrt{\frac{C}{I}}$$

Lec-8



$C = \frac{\pi G D_{wire}^4}{32L}$

$I = \frac{1}{2} M R_{disc}^2$

$= \frac{1}{8} M D_{disc}^2$

$\tau = -C\theta = I\frac{d^2\theta}{dt^2}$

$\frac{d^2\theta}{dt^2} = \ddot{\theta} = -\frac{C}{I}\theta$

$\downarrow$

ω^2

$= -\omega^2\theta$

$\omega = \sqrt{\frac{C}{I}}$

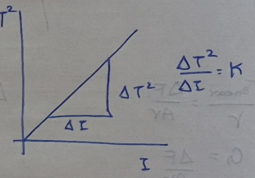
$\therefore T = 2\pi\sqrt{\frac{I}{C}}$

$T^2 = 4\pi^2\frac{I}{C}$

$T^2 \propto I$

$= KI$

	T	T^2	$I \approx I_0 + 2MR^2$
1	T_1	T_1^2	I_1
2	T_2	T_2^2	I_2
3			
	T_n	T_n^2	I_n



$\frac{\Delta T^2}{\Delta I} = K$

#ex $C_{steel} = 8\times10^{10}$ Pa

$D_{disc} = 10$ cm

$M_{disc} = 1$ kg

$I = \frac{1}{8}\times1\times0.1\times0.1 = \frac{0.01}{8} = \frac{1}{800}$

$C = \frac{\pi\times8\times10^{10}\times(10^{-3})^4}{4\times32\times L}$

$= \frac{\pi}{4L}\times10^6$

$T = \frac{2\pi}{2}\sqrt{\frac{\frac{1}{800}}{\frac{\pi}{4L}\times10^6}}$

$= \pi\sqrt{\frac{L}{8\pi\times10^8}}$

$= \frac{\pi}{2}\sqrt{\frac{L}{2\pi}}\times10^{-4}$

$= \frac{\pi\times10^{-4}}{2\sqrt{2\pi}}$

$= \frac{\pi\times10^{-4}}{5.01} = 0.2\pi\times10^{-4}$

ex: $C_{steel} \approx 8\times10^{10}$ Pa

$D_{disc} \approx 10\text{cm} = 0.1\text{m}$

$M_{disc} \approx 1$ Kg

$D_{thread} = 1\text{mm} = 10^{-3}$ m

$C = \frac{\pi\times8\times10^{10}\times(10^{-3})^4}{32\times4}$

$= \frac{\pi}{4}\times10^{-2}$

$= 0.78\times10^{-2}$

$\approx 8\times10^{-3}$

$I = \frac{1}{8}\times1\times0.1\times0.1$

$= \frac{1}{8}\times10^{-2}$

$\omega = \sqrt{\frac{C}{I}} \approx \frac{8}{\sqrt{10}} = 2.5$

$\nu = \frac{\omega}{2\pi} \approx \frac{2.5}{2\pi}$

$T = 2.5$ s

Q: 1-11

A = 5cm

$\nu = 1$ Hz

$\omega = 2\pi\nu$

$x(0) = 0$

$x = A\cos(\omega t + \alpha)$

a) $\omega = 2\pi\nu$

$= 2\pi$

$\alpha =$

$0 = 5\cos(\alpha)$

$\alpha = \frac{\pi}{2}$

b) At $t = \frac{8}{3}$ sec

$x = 5\cos\left(2\pi\times\frac{8}{3} + \frac{\pi}{2}\right)$

$= 5\cos\left(\frac{16\pi}{3} + \frac{\pi}{2}\right)$

$= 5\cos\left(\frac{32\pi+3\pi}{6}\right)$

$= 5\cos\left(\frac{35\pi}{6}\right)$

$\frac{dx}{dt} = -5\times\frac{2\pi}{3}\sin\left(\frac{35\pi}{6}\right)$

$\frac{d^2x}{dt^2} = \frac{dx}{dt} = \frac{dA\cos(\omega t+\alpha)}{dt}$

$= -A\sin(\omega t)$

$= -5\times2\pi\sin\left(2\pi\times\frac{8}{3}\right)$

$\frac{d^2x}{dt^2} = -A\omega^2\cos(\omega t)$

$= -5\times4\pi^2\cos\left(2\pi\times\frac{8}{3}\right)$

1-12

③ $\nu = 1 \quad x(0) = 1 \quad v(0) = 1$

Find ω, A, α and show that $x(t) = \sqrt{2}\sin\left(2\pi t + \frac{\pi}{4}\right)$

$x(t) = A\cos(\omega t + \alpha)$

$x'(t) = -A\omega\sin(\omega t + \alpha)$

$x(0) = 1$

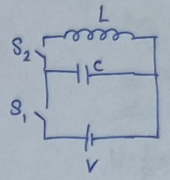
$1 = A\cos\alpha$

$1 = -A\omega\sin\alpha$

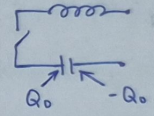
$2\pi\nu = \omega$

$\omega = 2\pi$

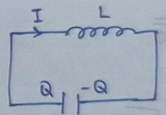
Lec - 9



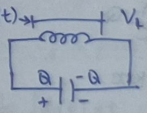
0. All switches are open.
1. Close S_1, charge on C, $Q = CV$
2. Open S_1



3. Close ~~S_2~~ S_2



$Q(0) = Q_0 = CV$

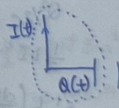


$V_L = L\frac{dI}{dt}$

$V_C = \frac{Q}{C}$

$V_L = V_C$

$L\frac{dI}{dt} = \frac{Q}{C}$



t: total charge - Q(t)

$t+\Delta t$: " $= Q(t+\Delta t) + I\Delta t$

Since these two are equal,

$Q(t+\Delta t) + I\Delta t = Q(t)$

$$\frac{Q(t+\Delta t)-Q(t)}{\Delta t} = -I$$

$$\frac{dQ}{dt} = -I$$

$$\therefore L\frac{dI}{dt} = \frac{Q}{C}$$

$$I = -\frac{dQ}{dt}$$

$$L\frac{d^2Q}{dt^2} = -\frac{1}{C}Q$$

$$\frac{d^2Q}{dt^2} = \frac{-1}{LC}Q$$

$$\omega^2 = \frac{1}{LC}$$

$$\omega = \frac{1}{\sqrt{LC}}$$

$$T = \frac{2\pi}{\omega}$$

$$T = 2\pi\sqrt{LC}$$

$$L\frac{d^2Q}{dt^2} = -\frac{1}{C}Q$$

$$m\frac{d^2x}{dt^2} = -kx$$

Variable	Comparison
x	Q
m	L
k	$1/C$
$U = \frac{1}{2}kq^2$	$Q^2/2C$
$K = \frac{1}{2}mv^2$	$\frac{1}{2}LI^2$
$= \frac{1}{2}m\left(\frac{dx}{dt}\right)^2$	$= \frac{1}{2}L\left(\frac{dQ}{dt}\right)^2$

Averages:

Avg KE, Avg U

$$x = A\sin\omega t$$

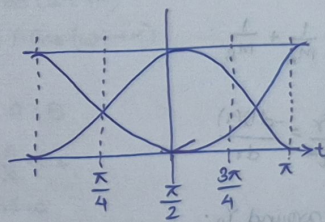
$$\dot{x} = A\omega\cos\omega t$$

$$U = \frac{1}{2}kx^2 = \frac{1}{2}kA^2\sin^2\omega t$$

$$\frac{k}{m} = \omega^2 \;;\; k = m\omega^2$$

$$U = \frac{1}{2}m\omega^2A^2\sin^2\omega t = \frac{1}{2}m\omega^2A^2\left(\frac{1-\cos 2\theta}{2}\right)$$

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2A^2\cos^2\omega t = \frac{m\omega^2A^2}{2}\left(\frac{1+\cos 2\theta}{2}\right)$$



Average Value Of a function

$$\langle f(t)\rangle_T = \frac{1}{T}\int_0^T f(t)\,dt$$

if $f(t+T) = f(t)$

then,

$$= \frac{1}{T}\int_{t_1}^{T+t_1} f(t)\,dt$$

$$\langle U\rangle = \frac{1}{T}\int_0^T \frac{1}{2}m\omega^2A^2\sin^2\omega t\,dt$$

$$= \frac{1}{2}\times\frac{1}{2\pi}\,\omega\cdot m\omega A^2\int_0^{2\pi}\sin^2\theta\,d\theta$$

$$= \frac{1}{2}m\omega^2A^2\times\frac{1}{2\pi}\int_0^{2\pi}\sin^2\theta\,d\theta = \frac{1}{2}$$

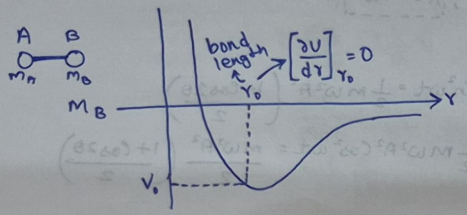
$$= \frac{1}{4}m\omega^2A^2$$

$$\langle U \rangle = \frac{1}{2} m\omega^2 A^2$$

$$\langle K \rangle = \frac{1}{4} m\omega^2 A^2$$

$$\langle K \rangle + \langle U \rangle = \frac{1}{2} m\omega^2 A^2$$

SHM in arbitary potentials having smooth minimum



$$\frac{1}{\mu} = \frac{1}{m_A} + \frac{1}{m_B}$$

Reduced mass

$$\mu \cdot \frac{d^2 r}{dt^2} = \frac{-dU(r)}{dr}$$

Taylor expansion around r_0.

$$U(r) = U(r_0) + (r - r_0)\left.\frac{dU}{dr}\right|^{0}_{r_0} + \frac{1}{2}(r - r_0)^2 \underbrace{\left.\frac{d^2U}{dr^2}\right|_{r_0}}_{K} + \dots$$

for small oscillations

$$\approx U(r_0) + \frac{1}{2}(r - r_0)^2 K$$

$$\approx U(r_0) + \frac{K}{2}(r - r_0)^2$$

$$\mu \frac{d^2 r}{dt^2} = \frac{-dU(r)}{dr}$$

$$= -K(r - r_0)$$

i.e. $$-\frac{dU}{dr} = -K(r - r_0)$$

define $(r - r_0) = x$

$$\frac{dx}{dt} = \frac{dr}{dt}, \quad \frac{d^2x}{dt^2} = \frac{d^2r}{dt^2}$$

i.e,

$$\mu \frac{d^2x}{dt^2} = -Kx$$

$$\omega^2 = \frac{K}{\mu}$$

$$\omega = \sqrt{\frac{K}{\mu}}$$

$$\omega = \sqrt{\frac{U''(r_0)}{\mu}}$$

$$X = A\cos(\omega t + \alpha)$$

$$r = r_0 + A\cos(\omega t + \alpha)$$

Example: Given A & B

$$U = \frac{A}{x} - Bx$$

find ω

$$\frac{dU}{dx} = \frac{-A}{x^2} - B \qquad \text{at } x_0 = \frac{-A}{x_0^2} - B$$

$$\frac{d^2U}{dx^2} = \frac{-2A}{x^3} \qquad \text{at } x_0 = \frac{-2A}{x_0^3}$$

Damped motion

$m\frac{dv}{dt} = -bv$

$F = 6\pi\eta r v$
$\propto v$
$= bv$

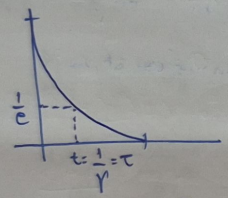
$\frac{dv}{dt} = \frac{-b}{m}v$

$\frac{-b}{m} = \gamma$

$\frac{dv}{dt} = -\gamma v$

$\int_{v_0}^{v} dv = -\gamma \int_0^t v\, dt$

$v = v_0 e^{-\gamma t} = v_0 e^{-t/\tau}$

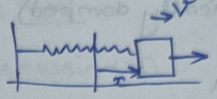


$\frac{dx}{dt} = v = v_0 e^{-\gamma t}$

$\int_{x_0}^{x} dx = v \int_0^t e^{-\gamma t} dt$

$K = \frac{1}{2}mv^2 = \frac{1}{2}mv_0^2 e^{-2\gamma t}$
$= K_0 e^{-2t/\tau}$

Damped Oscillator



$F_f = -bv = -b\frac{dx}{dt}$

$m\frac{d^2x}{dt^2} = \underbrace{-kx}_{f_{spring}} = \underbrace{-bv}_{f_{damping}}$

$\frac{d^2x}{dt^2} + \frac{b}{m}\frac{dx}{dt} + \underbrace{\frac{k}{m}}_{\omega_0^2} x = 0$

$\sqrt{\frac{k}{m}} = \omega_0 =$ natural freq

$\frac{b}{m} = \gamma =$ damping constant

$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$

$x = Ce^{\lambda t}$

H.W $x(t) = Ae^{-\frac{\gamma}{2}t}\cos(\omega t + \alpha)$

$\omega^2 = \omega_0^2 - \left(\frac{\gamma}{2}\right)^2$

$\omega = \sqrt{\omega_0^2 - \left(\frac{\gamma}{2}\right)^2} = \omega_0\left[1 - \left(\frac{\gamma}{2\omega_0}\right)^2\right]^{1/2}$

$= \omega_0\left[1 - \left(\frac{1}{2\omega_0\tau}\right)^2\right]^{1/2}$

$\langle K\rangle \approx \frac{1}{4}m\omega_0^2 A^2 e^{-\gamma t}$

$\langle U\rangle \approx \frac{1}{4}m\omega_0^2 A^2 e^{-\gamma t}$

$\langle \epsilon\rangle \approx \frac{1}{2}m\omega_0^2 A^2 e^{-\gamma t}$

H.W $P = \frac{d\langle\epsilon\rangle}{dt} = -\gamma\langle\epsilon\rangle = \langle Fv\rangle$

Damped Oscillator

(underdamped, overdamped, critically damped)

$$F = F_{spring} - F_{damp} = ma$$

$$= -kx - b\frac{dx}{dt} = m\frac{d^2x}{dt^2}$$

$$\frac{d^2x}{dt^2} + \frac{k}{m}x + \frac{b}{m}\frac{dx}{dt} = 0$$

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$$

$$x = Ae^{\lambda t}$$

$$\underbrace{(\lambda^2 + \gamma\lambda + \omega_0^2)}_{=0} \underbrace{Ae^{\lambda t}}_{\neq 0} = 0$$

$$\lambda^2 + \gamma\lambda + \omega_0^2 = 0$$

$$\left(\lambda + \frac{\gamma}{2}\right)^2 + \omega_0^2 - \left(\frac{\gamma}{2}\right)^2 = 0$$

Case 1

$$\omega_0 > \frac{\gamma}{2}$$

$$\left(\lambda + \frac{\gamma}{2}\right)^2 = -\left[\omega_0^2 - \left(\frac{\gamma}{2}\right)^2\right]$$

$$\lambda + \frac{\gamma}{2} = \pm i\sqrt{\omega_0^2 - \frac{\gamma^2}{4}}$$

$$x(t) = C_1 e^{-\frac{\gamma}{2}t + i\omega t} + C_2 e^{-\frac{\gamma}{2}t} e^{-i\omega t}$$

$$= e^{-\frac{\gamma t}{2}} \cos(\omega t + \alpha)$$

$$x(0) = x_0 \quad \dot{x}(0) = v(0) = v_0$$

$$x_0 = A\cos\alpha = x_0$$

$$x = Ae^{-\frac{\gamma t}{2}} \cos(\omega t + \alpha)$$

$$\dot{x} = -\frac{\gamma}{2} Ae^{-\frac{\gamma t}{2}} \cos(\omega t + \alpha) - \omega Ae^{-\frac{\gamma t}{2}} \sin(\omega t + \alpha)$$

~~to~~

$$\dot{x}(0) = -\frac{\gamma}{2} A\cos\alpha - \omega A\sin\alpha = v_0$$

Assume $v_0 = 0$

$$\frac{\gamma}{2}\cos\alpha + \omega\sin\alpha = 0$$

$$\tan\alpha = \frac{-\gamma}{2\omega}$$

$$\alpha = \tan^{-1}\left(\frac{-\gamma}{2\omega}\right)$$

$$= -\tan^{-1}\left(\frac{\gamma}{2\omega}\right)$$

$$x(0) = 0, \quad v(0) = v_0$$

$$A\cos\alpha = 0, \quad \alpha = \pm\frac{\pi}{2}$$

$$x(t) = Ae^{-\frac{\gamma t}{2}} \cos\left(\omega t - \frac{\pi}{2}\right)$$

$$= Ae^{-\frac{\gamma t}{2}} \sin\omega t$$

$$\dot{x} = -\frac{\gamma}{2} Ae^{-\frac{\gamma t}{2}}\left[-\frac{\gamma}{2}\sin\omega t + \omega\cos\omega t\right]$$

$$\cos\varphi = \frac{-\gamma/2}{\sqrt{\left(\frac{\gamma}{2}\right)^2 + \omega^2}} = \frac{-\gamma}{2\omega_0}$$

$$\sin\varphi = \frac{\omega}{\sqrt{\left(\frac{\gamma}{2}\right)^2 + \omega^2}} = \frac{\omega}{\omega_0}$$

$$\tan\varphi = \frac{\frac{\omega}{\omega_0}}{\frac{-\gamma}{2\omega_0}} = \frac{-2\omega}{\gamma}$$

$$\varphi = \tan^{-1}\left(\frac{-2\omega}{\gamma}\right)$$

$$\dot{x} = Ae^{-\gamma t}$$

$U = \frac{1}{2}kx^2 = \frac{1}{2}kA^2 e^{-\gamma t}\sin^2\omega t$

$\downarrow \omega^2 m$

$= \frac{1}{2}\omega^2 m A^2 e^{-\gamma t}\sin^2\omega t$

$\langle U\rangle \approx \frac{1}{4}\omega^2 m A^2 e^{-\gamma t}$ [assuming $e^{-\gamma t} \approx$ const]

$v = \omega_0 A e^{-\frac{\gamma t}{2}}\sin(\omega t + \phi - \varphi_1)$

$\left\langle \frac{1}{2}mv^2 \right\rangle \approx \frac{1}{2}m\omega_0^2 A^2 e^{-\frac{\gamma t}{2}} \langle \sin^2(\omega t - \varphi_1)\rangle$

$\approx \frac{1}{2}m\omega_0^2 A^2 e^{-\gamma t}$

$\langle E\rangle = \langle U\rangle + \langle K\rangle$

$\approx \frac{1}{2}m\omega_0^2 A^2 e^{-\gamma t}$

$\frac{d\langle E\rangle}{dt} = -\gamma\langle E\rangle$ = Power

Power dissipated $= \gamma\langle E\rangle$

Quality factor: $\frac{2\pi \times E_{stored}}{E_{loss\ in\ one\ T}}$

$= \frac{2\pi\langle E\rangle}{P\cdot T} = \frac{2\pi\langle E\rangle}{P/F}$

$Q = \frac{\omega\langle E\rangle}{P} = \frac{\omega}{\gamma} \approx \frac{\omega_0}{\gamma}$

If $\boxed{\omega_0 > \frac{\gamma}{2}}$

$x = e^{-\frac{\gamma}{2}}\left[C_1 e^{kt} + C_2 e^{-kt}\right]$

$k = \sqrt{\left(\frac{\gamma}{2}\right)^2 - \omega_0^2}$, $\omega_0 < \frac{\gamma}{2}$

Critically damped $\frac{\gamma}{2} = \omega_0$

$x(t) = (A+Bt)e^{-\frac{\gamma t}{2}}\sin\omega t$.

13-07-26 Lec-12

Problems: Chapter 3 : 12, 13, 14, 15, 16

$* \ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$

$\omega_0 = 2$ rad/s

$\gamma = 4$ Ns/mK

$= 4s^{-1}$

$x(0) = 0$

$\dot{x}(0) = 1$

$\lambda^2 + \gamma\lambda + \omega_0^2 = 0$

$\lambda^2 + 4\lambda + 4 = 0$

$\lambda + 2\lambda + 2\lambda + 4 = 0$

$\lambda(\lambda+2) + 2(\lambda+2) = 0$

$(\lambda+2)^2 = 0$

$\lambda_1, \lambda_2 = -2$

Gen soln:

$x(t) = C_1 e^{\lambda t} + C_2 t e^{\lambda t}$

$\dot{x}(t) = C_1\lambda e^{\lambda t} + C_2\left[t\lambda e^{\lambda t} + e^{\lambda t}\right]$

$= C_1\lambda e^{\lambda t} + C_2 t\lambda e^{\lambda t} + C_2 e^{\lambda t}$

$= (C_1\lambda + C_2 t\lambda + C_2)e^{\lambda t}$

$x(0) = C_1$

$0 = C_1$

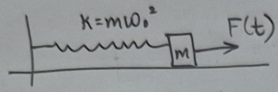
$\dot{x}(0) = 1$

$1 = 0 + C_2$

$C_2 = 1$

H.W: $x(0) = 1$, $\dot{x}(0) = 0$ $* \ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$

Driven Oscillations, Resonance (Ch-4)



Undamped case:

$$m\ddot{x} = -kx + F_0 \cos\omega t$$

$$\ddot{x} = -\frac{k}{m}x + \frac{F_0}{m}\cos\omega t \longrightarrow f_0 \text{ (let)}$$

$$\ddot{x} = -\omega_0^2 x + f_0 \cos\omega t$$

$$\ddot{x} + \omega_0^2 x - f_0 \cos\omega t = 0$$

Particular sol^n:

Choose $x = A\cos(\omega t + \alpha)$

$$\dot{x} = -A\omega \sin(\omega t + \alpha)$$

$$\ddot{x} = -A\omega^2 \cos(\omega t + \alpha)$$

$$-A\omega^2 \cos(\omega t + \alpha) + \omega_0^2 x + A \cancel{\omega^2} \cos(\omega t + \alpha) = f_0 \cos\omega t$$

$$(-\omega^2 + \omega_0^2) A\cos(\omega t + \alpha) = f_0 \cos\omega t$$

Let $f_0 > 0$

$$A\cos(\omega t + \alpha) = \frac{f_0}{\omega_0^2 - \omega^2}\cos\omega t$$

for $\omega < \omega_0$,

$$A = \frac{f_0}{\omega_0^2 - \omega^2}$$

$\cos(\omega t + \alpha) = \cos\omega t \Rightarrow \alpha = 0$

(max oscillation in phase with driven oscillation)

for $\omega > \omega_0$,

$$A = \frac{f_0}{\omega^2 - \omega_0^2}$$

$\cos(\omega t + \alpha) = -\cos\omega t \Rightarrow \alpha = \pi$

(max oscillation out a phase by π with driven oscillation)

$$A = \frac{f_0}{|\omega^2 - \omega_0^2|}$$

$$\alpha = \begin{cases} 0, & \omega < \omega_0 \\ \pi, & \omega > \omega_0 \end{cases}$$

$$\ddot{x} + \omega_0^2 x = f_0 \cos\omega t \quad -①$$

$$\ddot{y} + \omega_0^2 y = f_0 \sin\omega t \quad -②$$

multiply i to eq^n ②

$$i\ddot{y} + i\omega_0^2 y + i f_0 \sin\omega t \quad -③$$

add ① & ③

$$\ddot{z} + \omega_0^2 z = f_0 e^{i\omega t}$$

$$z = Ae^{i(\omega t + \alpha)}$$

$$(-\omega^2 + \omega_0^2) Ae^{i(\omega t + \alpha)} = f_0 e^{i\omega t}$$

$$(\omega_0^2 - \omega^2) Ae^{i\alpha} = f_0$$

$\omega < \omega_0$,

$$A = \frac{f_0}{|\omega^2 - \omega_0^2|},$$

$$\alpha = \begin{cases} 0, & \omega < \omega_0 \\ \pi, & \omega > \omega_0 \end{cases}$$

$$z = Ae^{i(\omega t + \alpha)}$$

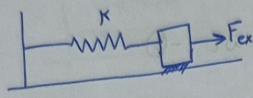
$$x = \text{Real}[z]$$

$$= A\cos(\omega t + \alpha)$$

$$= \begin{cases} A\cos\omega t, & \omega < \omega_0 \\ -A\cos\omega t, & \omega_0 > \omega \end{cases}$$

Lec-13

Driven ~~&~~ damped oscillation



$$m\frac{d^2x}{dt^2} = F = -Kx - b\frac{dx}{dt} + F_0\cos\omega t$$

$$\ddot{x} = -\frac{K}{m}x - \frac{b}{m}\frac{dx}{dt} + \frac{F_0}{m}\cos\omega t$$

ω_0 = natural angular freq $= \sqrt{\frac{K}{m}}$

γ = damping factor $= \frac{b}{m} = \frac{1}{\tau}$

$f_0 = \frac{F_0}{m}$

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = f_0\cos\omega t \quad -(1)$$

$x = x(t)$

STEADY STATE MOTION OF DAMPED AND DRIVEN OSC

$x(t) = A\cos(\omega t - \delta)$ (δ: phase lag)

let $\ddot{y} + \gamma\dot{y} + \omega_0^2 y = f_0\sin\omega t$ (another oscillator in y driven by $f_0\sin\omega t$) $-(2)$

multiply i to (2)

$$i(\ddot{y} + \gamma\dot{y} + \omega_0^2 y) = i(f_0\sin\omega t) \quad -(2')$$

eqn 1+2'

$$\ddot{x} + i\ddot{y} + \gamma(\dot{x} + i\dot{y}) + \omega_0^2(x + iy) = f_0 e^{i\omega t}$$

$$\ddot{z} + \gamma\dot{z} + z\omega_0^2 = f_0 e^{i\omega t}$$

$\omega_d = \sqrt{\omega_0^2 - \left(\frac{\gamma}{2}\right)^2}$: (freq of underdamped oscillator).

Steady state $= z = \tilde{A}e^{i\omega t}$ $\qquad \tilde{A} = Ae^{-i\delta}$

$$= Ae^{-i\delta}e^{i\omega t}$$

$$z = Ae^{i(\omega t - \delta)}$$

$$\dot{z} = i\omega Ae^{i(\omega t - \delta)}$$

$$\ddot{z} = i^2\omega^2 Ae^{i(\omega t - \delta)}$$

$$= -\omega^2 Ae^{i(\omega t - \delta)}$$

$$(-\omega^2 A + i\omega\gamma A + \omega_0^2 A)e^{i(\omega t - \delta)} = f_0 e^{i\omega t}$$

$$\left[(\omega_0^2 - \omega^2)A + i\omega\gamma A\right] = f_0 e^{i\delta}$$

$$(\omega_0^2 - \omega^2)A + i\omega\gamma A = f_0[\cos\delta + i\sin\delta]$$

$$\boxed{(\omega_0^2 - \omega^2)A = f_0\cos\delta \qquad \omega\gamma A = f_0\sin\delta}$$

$$f_0^2 = \left((\omega_0^2 - \omega^2)A\right)^2 + (\omega\gamma A)^2$$

$$f_0^2 = (\omega_0^2 - \omega^2)^2 A^2 + \omega^2\gamma^2 A^2$$

$$f_0^2 = \left((\omega_0^2 - \omega^2)^2 + \omega^2\gamma^2\right)A^2$$

$$A = \frac{f_0^2}{(\omega_0^2 - \omega^2)^2 + \omega^2\gamma^2}$$

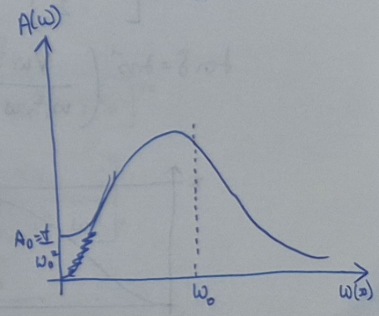
$$\boxed{A = \frac{f_0}{\left[(\omega_0^2 - \omega^2)^2 + (\gamma\omega)^2\right]^{1/2}}}$$

~~tan~~

$$\boxed{\tan\delta = \frac{\gamma\omega}{\omega_0^2 - \omega^2}}$$

~~At~~

$A(\omega_0) = \frac{f_0}{\gamma\omega_0}$



$$A'(\omega) = \frac{2f_0 \times (\omega_0^2 - \omega) 2\omega - \gamma^2 \omega}{(\omega_0^2 - \omega^2) + (\gamma\omega)^2}$$

$$= \frac{2f_0 \left[(\omega^2 - \omega_0^2) \cdot 2\omega + \gamma^2 \omega\right]}{2\left[(\omega_0^2 - \omega^2) + (\gamma\omega)^2\right]}$$

$A'(\omega) = 0$ for max;

$$2f_0 \left[(\omega^2 - \omega_0^2) 2\omega + \gamma^2 \omega\right] = 0$$

$$(\omega^2 - \omega_0^2) 2\omega + \gamma^2 \omega = 0$$

$$2\omega \left[(\omega^2 - \omega_0^2) + \frac{\gamma^2}{2}\right] = 0$$

$\omega = 0$ or $\omega^2 - \omega_0^2 + \frac{\gamma^2}{2} = 0$

$$\omega = \sqrt{\omega_0^2 - \frac{\gamma^2}{2}}$$

$$= \omega_0 \left[1 + (\gamma/\omega_0)^2\right]^{1/2}$$

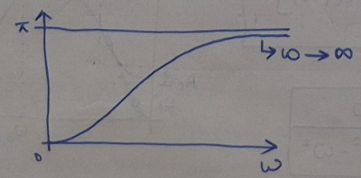
$$\frac{d}{d\omega}\left[(\omega^2 - \omega_0^2) + \gamma^2\omega^2\right] =$$

$$2(\omega^2 - \omega_0^2) 2\omega + \gamma^2 \cdot 2\omega$$

$$= 4\omega \left[\omega^2 - \left(\omega_0^2 - \frac{\gamma^2}{2}\right)\right] = 0$$

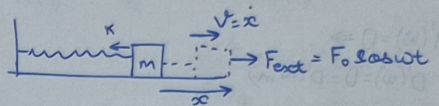
$$\omega = \left[\omega_0^2 - \frac{\gamma^2}{2}\right]^{1/2}$$

$$\tan\delta = \tan^{-1}\left(\frac{\gamma\omega}{\omega_0^2 - \omega^2}\right) \quad 0 \le \delta \le \pi$$



Lec 14

Driven & Damped Oscillation



$$\vec{F}_f = -b\dot{x}$$

$$F_{spring} = -Kx$$

$$m\ddot{x} = -Kx - b\dot{x} + F_0\cos\omega t$$

$$\ddot{x} + \frac{b}{m}\dot{x} + \frac{K}{m}x = f_0\cos\omega t$$

$$\ddot{x} + \gamma\dot{x} + \omega^2 x = f_0\cos\omega t$$

$z = x + iy$, $\vec{F} = f_0 e^{i\omega t}$

$$\ddot{z} + \gamma\dot{z} + \omega_0^2 z = f_0 e^{i\omega t}$$

$x = \text{Real}[z]$

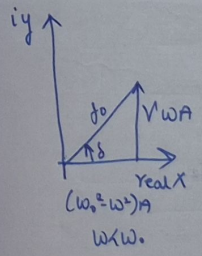
$z = Ae^{(\omega t - \delta)}$

$\dot{z} = i\omega A e^{(i\omega t - \delta)}$

$\ddot{z} = i^2\omega^2 A e^{(i\omega t - \delta)}$

$= -\omega^2 A e^{(i\omega t - \delta)}$

$$(\omega_0^2 - \omega^2)A + i\gamma\omega A = f_0 e^{i\delta} = f_0(\cos\delta + i\sin\delta)$$



$$A(\omega) = \frac{f_0}{\left[(\omega_0^2 - \omega^2) + \gamma^2\omega^2\right]^{1/2}}$$

$$\delta(\omega) = \tan^{-1}\left[\frac{\gamma\omega}{\omega_0^2 - \omega^2}\right]$$

$$A(\omega_0) = \frac{f_0}{\gamma\omega_0}$$

$$A(0) = \frac{f_0}{\omega_0^2}$$

$$D(\omega) = \left[(\omega^2 - \omega_0^2)^2 + r^2\omega^2\right]$$

$$A(\omega) = \frac{f_0}{\sqrt{D(\omega)}}$$

$$A'(\omega) = 0 \Rightarrow$$
$$D'(\omega) = 0 = D'(\omega_m)$$
$$\omega_m^2 = \omega_0^2 - \frac{r^2}{2}$$

$A(\omega)$ is max at $\omega_m^2 = \omega_0^2 - \frac{r^2}{2}$

damped natural freq $\omega_d^2 = \omega_0^2 - \frac{r^2}{4}$

$$D(\omega) = (\omega - \omega_m)^2 + r^2\omega_d^2$$

$$D(\omega_m) = r^2\omega_d^2$$

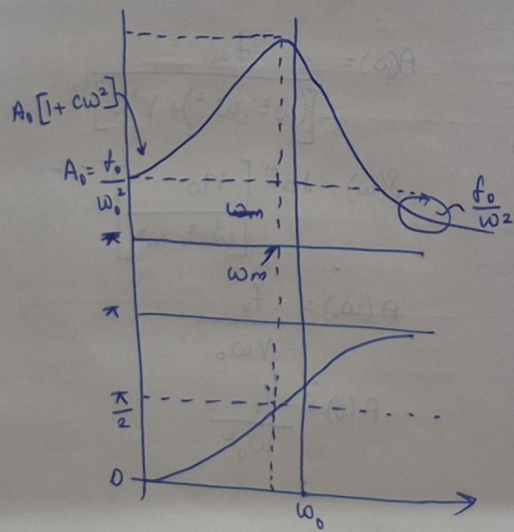
$$A(\omega_m) = \frac{f_0}{\sqrt{D(\omega_m)}} = \frac{f}{r\omega_d}$$

$$\omega_d = \sqrt{\omega_0^2 - \frac{r^2}{4}} = \omega_0\sqrt{1 - \frac{r^2}{4\omega_0^2}}$$

Q = "Quality factor" $= \frac{\omega_0}{r}$

$$\omega_d = \omega_0\left[1 - \frac{1}{4Q^2}\right]^{1/2}$$

$$\omega_m = \omega_0\left[1 - \frac{1}{2Q^2}\right]^{1/2}$$



$$\frac{\omega_0}{r} = Q \Rightarrow r = \frac{\omega_0}{Q}$$

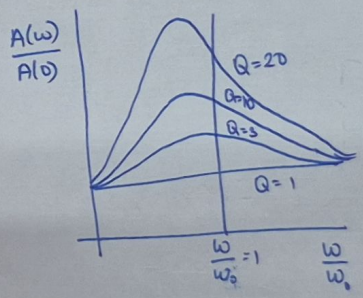
$$\frac{A(\omega_m)}{A(0)} = \frac{A_m}{A_0} = \frac{f_0/r\omega_d}{f_0/\omega_0^2}$$

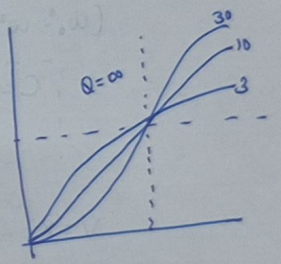
$$= \frac{\omega_0^2}{r\omega_d} = \frac{\frac{\omega_0}{r}\times\omega_0}{\omega_0\left[1 - \frac{1}{4Q^2}\right]^{1/2}}$$

$$= \frac{Q}{\left[1 - \frac{1}{4Q^2}\right]^{1/2}}$$

$$\delta(\omega) = \tan^{-1}\left[\frac{r\omega}{\omega_0^2\left[1 - \frac{\omega^2}{\omega_0^2}\right]}\right]$$

$$= \tan^{-1}\left[\frac{\omega/\omega_0}{Q\left[1 - \frac{\omega^2}{\omega_0^2}\right]}\right]$$





Lec 15

Power Absorbed by driven oscillator

Undamped osc

$$\frac{d^2x}{dt^2} + \omega_0^2 x = \frac{F_0}{m}\cos\omega t$$

$$x = A\cos(\omega t + \alpha)$$

$$A = \begin{cases} \dfrac{f_0}{m(\omega_0^2 - \omega^2)}, & \omega < \omega_0 \\ \dfrac{f_0}{m(\omega^2 - \omega_0^2)}, & \omega > \omega_0 \end{cases}$$

$$\alpha = \begin{cases} 0, & \omega < \omega_0 \\ \pi, & \omega > \omega_0 \end{cases}$$

$$x = C\cos\omega t$$

$$(\omega_0^2 - \omega^2)C = \frac{F_0}{m}$$

$$C = \frac{F_0}{m(\omega_0^2 - \omega^2)}$$

$$V = \dot{x} = -\omega C \sin\omega t$$

$$\dot{W} = P = F\frac{dx}{dt} = Fv$$

$$= F_0\cos\omega t\,(-\omega C)\sin\omega t$$

Damped and driven Oscillator

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = F_0\cos\omega t$$

$$x = A\cos(\omega t - \delta)$$

$$A = \frac{f_0}{\left[(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2\right]^{1/2}} = \frac{LT^{-2}}{T^{-2}} = L$$

$$z = Ae^{i(\omega t - \delta)} \qquad RHS = F_0 e^{i\omega t}$$

$$(\omega_0^2 - \omega^2)A + i\gamma\omega A = f_0(\cos\delta + i\sin\delta)$$

$$(\omega_0^2 - \omega^2)A = f_0\cos\delta$$

$$\gamma\omega A = f_0\sin\delta$$

$$\dot{W} = P = Fv$$

$$V = -\omega A\sin(\omega t - \delta)$$

$$= -V_0(\omega)\sin(\omega t - \delta)$$

$V_0(\omega)$ → Velocity amplitude $= \omega A$

$$P = -F_0 V_0\cos\omega t\sin(\omega t - \delta)$$

$$= -F_0 V_0[\cos\omega t\sin\omega t\cos\delta - \cos\omega t\cdot\cos\omega t\sin\delta]$$

$$\overline{P} = F_0 V_0\left[\underbrace{\cos^2\omega t}\sin\delta - \frac{1}{2}\underbrace{\sin 2\omega t}_{0}\cos\delta\right]$$

$$\bar{P} = \frac{1}{2} F_0 v_0 \sin\delta = \frac{1}{2} m\omega r A v_0$$

$$\boxed{\bar{P} = \frac{1}{2} m\omega^2 A^2 r}$$

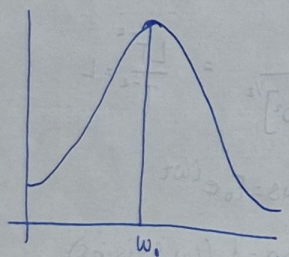
$$P = \frac{1}{2} m\omega^2 r \frac{f_0^2}{(\omega_0^2 - \omega^2)^2 + r^2\omega^2}$$

$$= \frac{1}{2} m\omega^2 r \frac{f_0^2}{\omega_0^4 \omega^2 \left(\frac{\omega_0}{\omega} - \frac{\omega}{\omega_0}\right)^2 + r^2}$$

$$= \frac{F_0^2 r}{2m\omega_0^2} \frac{1}{\left(\frac{\omega_0}{\omega} - \frac{\omega}{\omega_0}\right)^2 + \frac{1}{Q^2}}$$

$$\boxed{Q = \frac{\omega_0}{r}}$$

$$P_{max} = \frac{F_0^2 r \omega_0^2}{2m\omega_0^2 r^2} = \frac{F_0^2}{2mr}$$



$$\omega = \omega_0 + \Delta\omega$$

$$= \omega_0 \left[1 + \frac{\Delta\omega}{\omega_0}\right]$$

$$= \omega_0 [1 + \eta]$$

$$\frac{\omega_0}{\omega} = \frac{1}{1+\eta} \approx 1 - \eta + \frac{\eta^2}{2} + \cdots$$

$$\frac{\omega}{\omega_0} = \frac{1}{1+\eta}$$

$$\frac{\omega_0}{\omega} - \frac{\omega}{\omega_0} = \left(1 - \eta + \frac{\eta^2}{2} + \cdots\right)$$

$$\frac{\omega_0}{\omega} - \frac{\omega}{\omega_0} = -2\eta + \frac{\eta^2}{2}$$

$$= -2\eta \left(1 - \frac{\eta}{4} + \cdots\right)$$

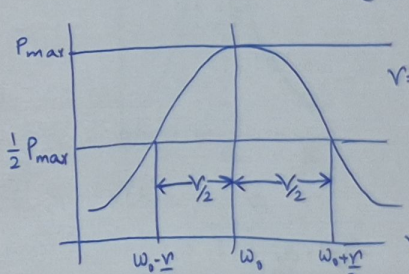
$$\left(\frac{\omega_0}{\omega} - \frac{\omega}{\omega_0}\right)^2 = 4\eta^2 \left(1 - \frac{\eta}{4} + \cdots\right)^2$$

$$= 4[\eta^2 + O(\eta^3)]$$

$$\approx 4\eta^2 = 4(\omega - \omega_0)^2/\omega_0^2$$

$$\bar{P}(\omega) = \frac{F_0^2 r}{2m\omega_0^2} \frac{1}{4\frac{(\omega - \omega_0^2)}{\omega_0^2} + \frac{r^2}{\omega_0^2}}$$

$$\bar{P}(\omega) = \frac{F_0 r}{8m} \frac{1}{(\omega - \omega_0)^2 + \left(\frac{r}{2}\right)^2}$$



full width with half max

$r = (\Delta\omega)_{fwhm}$ $\quad Q = \frac{\omega_0}{r}$

$r = \frac{\omega_0}{Q}$

Resonance width $= \frac{\omega_0}{Q}$

Lec-16

Free damped Oscillation

$$\epsilon(t) = \epsilon_0 e^{-rt}$$

$$\frac{d\epsilon}{dt} = -r\epsilon_0 \Rightarrow \frac{\Delta\epsilon}{\Delta t} \approx -r\epsilon_0$$

$$\frac{\Delta\epsilon}{\epsilon} \approx -r\Delta t$$

$$\left|\frac{\Delta\epsilon}{\epsilon}\right| \sim rt = r\frac{2\pi}{\omega_0} = 2\pi$$

$$Q = \frac{2\pi\epsilon}{\Delta\epsilon \text{ (cycle)}}.$$

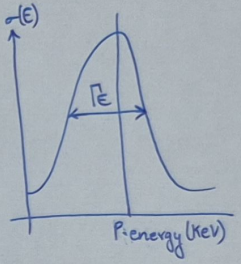
LCR- Circuit

Power decipated : Assignment.

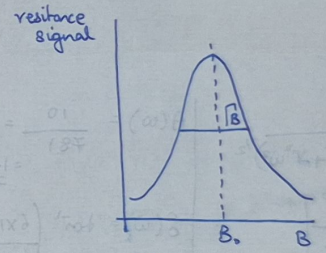
~~Resonance~~

Resonance in the yield of γ-rays.

$$P + {}^{19}F \longrightarrow {}^{2}Ne^{*} + \gamma$$



$$\sigma(E) = \frac{\sigma(E_0)}{\frac{(E-E_0)^2}{4\Gamma^{-2}} + 1}$$



$$V(B) = \frac{V_0}{1 + \frac{(B-B_0)^2}{4\Gamma_B^2}}$$

Superposition principle

Linear ODE such as,

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$$

* If $x_1(t)$, $x_2(t)$ are solutions
then $x = C_1 x_1(t) + C_2 x_2(t)$ is a solution
* If $x_1(t)$ is a solution where $x_1(0) = x_{10}$, $\dot{x}_1(0) = v_{10}$
$x_2(t)$ is " " " $x_2(0) = x_{20}$, $\dot{x}_2(0) = v_{20}$

then if $x(0) = x_{10} + x_{20}$; $\dot{x}(0) = \dot{x}_{10} + \dot{x}_{20}$

$$x(t) = x_1(t) + \dot{x}_1(t)$$

Lec-17

4-10

a) $F = -bv$

$P = Fv$

$P = -bv^2$

b) $x = A\cos(\omega t - \delta)$

$\dot{x} = -A\omega\sin(\omega t - \delta)$

$v^2 = A^2\omega^2\sin^2(\omega t - \delta)$

$P = -bA^2\omega^2\sin^2(\omega t - \delta)$

$\langle P \rangle = \frac{1}{2} bA^2\omega^2$

$\frac{b}{m} = \gamma$

$b = m\gamma$

$\langle P \rangle = \frac{1}{2} m\gamma A^2\omega^2$

4-11 $A(\omega) = \frac{F_0}{m} \frac{1}{\left((\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2\right)^{1/2}}$

$\delta(\omega) = \tan^{-1}\left[\frac{\gamma\omega}{\omega_0^2 - \omega^2}\right]$

$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{80}{0.2}} = 20\,s^{-1}$

~~$b =$~~ $\gamma = \frac{b}{m} = \frac{4}{0.2} = 20\,s^{-1}$

$\frac{F_0}{m} = \frac{\alpha}{0.2} = 10$

$(\omega_0^2 - \omega^2) = (5\times100)^2$

$(\gamma\omega)^2 = (6\times100)^2$

$\sqrt{(\omega_0^2 - \omega^2) + (\gamma\omega)^2} = \sqrt{(5\times100)^2 + (6\times100)^2}$

$= \sqrt{(5^2 + 6^2)100^2}$

$= 100 \times 7.81$

$= 781$

$A(\omega) = \frac{10}{781} = 0.0128\,m$

$= 1.28\,cm$

$\delta(\omega) = \tan^{-1}\left(\frac{6\times100}{(5\times100)^2}\right)$

$\delta = -\tan^{-1}\left(\frac{6}{5}\right)$

$\delta = -50.19° + \pi$

$\delta = 129.81°$

δ always btw 0 & π

$\delta = pi - 0.876$

$= 2.26\,rad$

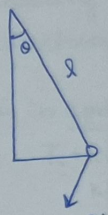
Lec-18

Problems 4-12, 4-13, 4-14, 4-17

4-2

extra: 4-5, 4-6

4-5



$-mg\sin\theta$

$-mg\sin\theta = -mg\frac{x}{\ell}$

$m\ell\ddot{\theta} = -mg\frac{x}{\ell}$

$\ell\theta \approx \ell\tan\theta \approx x$

$m\ddot{x} \approx -\frac{mg}{\ell}x$

$m\ddot{x} = -\frac{mg}{\ell}(x - \gamma)$

$\ddot{x} + \gamma\dot{x} + \frac{g}{\ell}x = \frac{g}{\ell}\gamma \Rightarrow \gamma\cos\omega t$

$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = \omega_0^2\gamma\cos\omega t$

$\frac{F_0}{m} = \omega_0^2\gamma$

Lec-19

$F = -Kx$

$ma = -Kx$

$K = \frac{-ma}{x}$

$m = 0.0001\,kg$

$x = 0.43785\,m$

$a = -1.7514\,cm/s^2$

$\omega = -0.04$

$A = \sqrt{(x_0)^2 + \left(\frac{V_0}{\omega}\right)^2}$

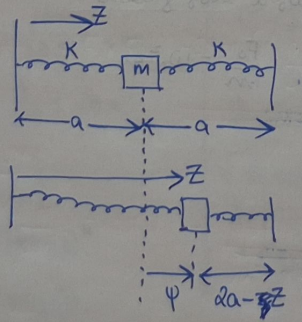
$= \sqrt{(0.43785)^2 + \left(\frac{0.1}{0.2}\right)^2}$

$= 0.664$

Lec-20

Two springs Mass System

Berkeley Physics Vol 3



M

a_0 equilibrium length.

In equm

$T_0 = K(a - a_0)$

assuming $a_0 < a$.

Pos of M

$Z = a + \psi$

ψ = displacement

$\dot{Z} = \dot{\psi}$ = velocity

$\ddot{Z} = \ddot{\psi}$ = acceleration

Tension in spring 1

$T_1 = K(Z - a_0)$

$= K(a + \psi - a_0)$

Restoring force from spring ①

$F_1 = -T_1$

Spring 2

$T_2 = K(2a - Z - a_0)$

$= K(a - \psi - a_0)$

Restoring force from spring ②

$F_2 = T_2$

E.o.M :

$M\ddot{\psi} = F_1 + F_2$

$= -T_1 + T_2$

$= -2K\psi$

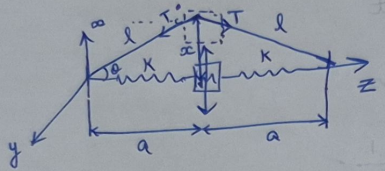
$\ddot{\psi} = \frac{-2K}{M}$, $\psi = -\omega^2\psi$

$\omega^2_{Long} = \frac{2K}{M}$

* Independent of a_0,

$a - |\psi_{max}| > a_0$

Transverse Oscillations



Let M get displaced along the x direction:

$$l = \sqrt{a^2+x^2}$$

T in string 1 $T_1 = K(l-a_0)$

F_{res} in string 1 $F_1 = -T_1 \sin\theta = -\frac{T_1 x}{l} = -\frac{K(l-a_0)x}{l}$

~~T in string 2 T_2 = def~~ $= -Kx\left(1-\frac{a_0}{l}\right)$

T in string 2 $T_2 = K(l-a_0)$

F_{res} in strin 2 $F_2 = -T_2 \sin\theta$

$= -Kx\left(1-\frac{a_0}{l}\right)$

$$F_{Net_{res}} = F_1 + F_2 = -2Kx\left(1-\frac{a_0}{l}\right)$$

EoM:

$$M\frac{d^2x}{dt^2} = -2Kx\left(1-\frac{a_0}{\sqrt{a^2+x^2}}\right)$$

Approximations:

1) Slinky approximation:

$a_0 \ll a$

$M\ddot{x} = -2Kx$

$\ddot{x} = -\frac{2Kx}{M}$

$\omega^2 = \frac{2K}{M}$

2) Small oscillation approximation

$$l = \sqrt{a^2+x^2} = a\left[1+\frac{x^2}{a^2}\right]^{1/2}$$

for small oscillation $|x| \ll a$, or

$\frac{a_0}{l} = \frac{a_0}{a}(1+\varepsilon)^{1/2}$; $\varepsilon = \frac{|x|}{a} \ll 1$

$\frac{a_0}{l} = \frac{a_0}{a}\left(1-\frac{1}{2}\frac{x^2}{a^2}+\cdots\right)$

$M\ddot{x} = -2Kx\left[1-\frac{a_0}{a}\left(1-\frac{1}{2}\frac{x^2}{a^2}+\cdots\right)\right]$

$= -2K\left(1-\frac{a_0}{a}\right)x$

$[l \approx a]$ linear aprox.

$\approx -\frac{2K}{a}(a-a_0)x$

$M\ddot{x} = -\frac{2T_0}{a}x$

$\ddot{x} = -\frac{2T_0}{Ma}x$

$\Rightarrow \omega_{Tr}^2 = \frac{2T}{Ma} = \frac{2K}{M}\left(1-\frac{a_0}{a}\right)$

$\omega_L^2 = \frac{2K}{M}$

$$\boxed{\frac{\omega_L}{\omega_T} = \frac{1}{\left(1-\frac{a_0}{a}\right)^{1/2}}}$$

Simplest system with two degree of freedom

For small x, y

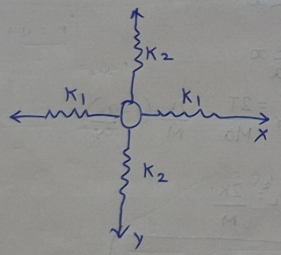
x-mode osc: $M\ddot{x} = -2Kx \Rightarrow \omega_1^2 = \frac{2K_1}{M}$

y-mode osc: $M\ddot{y} = -2K_2 y \Rightarrow \omega_2^2 = \frac{2K_2}{M}$

} Two modes

Lec 21

Coupled Oscillation with two degree of freedom



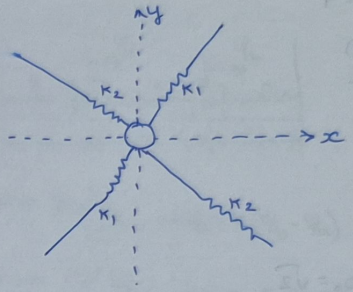
$$\ddot{x} = -\frac{2K_1}{M}x \Rightarrow \omega_1^2 = \frac{2K_1}{M}$$

$$\ddot{y} = -\frac{2K_2}{M}y \Rightarrow \omega_2^2 = \frac{2K_2}{M}$$

$$x = A_1 \cos(\omega_1 t + \psi_1)$$

$$y = A_2 \cos(\omega_2 t + \psi_2)$$

$$\vec{r} = \hat{i} A_1 \cos(\omega_1 t + \psi_1) + \hat{j} A_2 \cos(\omega_2 t + \psi_2)$$



$$\ddot{x} = -a_{11}x - a_{12}y$$

$$\ddot{y} = -a_{21}x - a_{22}y$$

$$a_{12} = a_{21}$$

Anormal Mode

* Every degree of freedom motion ~~oscillates with~~
- oscilates with same freq.
- oscilates with same phase.
- equilibrium at the same time.

$$X = A\cos(\omega t + \psi)$$

$$Y = B\cos(\omega t + \psi)$$

Example

$$\ddot{x} = -4x - 2y$$

$$\ddot{y} = -2x - 4y$$

$$x = A\cos(\omega t + \psi) \qquad y = B\cos(\omega t + \psi)$$

$$\ddot{x} = -\omega^2 x \;, \quad \ddot{y} = -\omega^2 y$$

$$-\omega^2 x = -4x - 2y$$

$$-\omega^2 y = -2x - 4y$$

$$(4 - \omega^2)x + 2y = 0 \qquad \vdots \qquad (4-\omega^2)y + 2x = 0$$

$$(4-\omega^2)x = -2y \qquad\qquad (4-\omega^2)y = -2x$$

$$\frac{y}{x} = \frac{4-\omega^2}{-2} = \frac{\omega^2 - 4}{2} \qquad\qquad \frac{y}{x} = \frac{-2}{4-\omega^2} = \frac{2}{\omega^2 - 4}$$

$$\frac{\omega^2-4}{2} = \frac{2}{\omega^2-4}$$

$$(\omega^2-4)^2 - 2^2 = 0$$

$$\omega^2-4 = \sqrt{2^2}$$

$$\omega^2-4 = \pm 2$$

$$\omega^2 = 4 \pm 2$$

$$\omega^2 = 6 \text{ or } 2$$

$$\omega_1 = \sqrt{6} \qquad \omega_2 = \sqrt{2}$$

$$\frac{y}{x} = \frac{B\cos(\omega t+\psi)}{A\cos(\omega t+\psi)}$$

$$\left(\frac{y}{x}\right)_{\text{mode 1}} = \left(\frac{B}{A}\right)_{\text{mode 1}} = \frac{B_1}{A_1}$$

$$\frac{B_1}{A_1} = \left(\frac{\omega^2-4}{2}\right)_{\text{mode 1}}$$

$$\frac{B_1}{A_1} = \frac{\omega_1^2-4}{2} = \frac{2-4}{2} = -1$$

$$B_1 = -A_1 \quad (\text{for mode 1, } \omega_1^2 = 2)$$

$$\left(\frac{y}{x}\right)_{\text{mode 2}} = \frac{B_2}{A_2} = \left(\frac{2}{\omega^2-4}\right) = \frac{2}{2} = 1$$

$$B_2 = A_2 \quad (\text{for mode 2, } \omega_2^2 = 6)$$

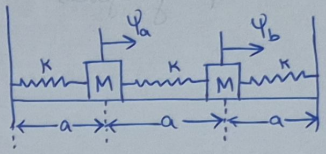
mode 1: $X_1 = A_1\cos(\omega_1 t+\psi_1)$, $Y_1 = -A_1\cos(\omega_1 t+\psi_1)$

mode 2: $X_2 = A_2\cos(\omega_2 t+\psi_2)$, $Y_2 = A_2\cos(\omega_2 t+\psi_2)$

General sol^n: $X = A_1\cos(\omega_1 t+\psi_1) + A_2\cos(\omega_2 t+\psi_2)$

$Y = -A_1\cos(\omega_1 t+\psi_1) + A_2\cos(\omega_2 t+\psi_2)$

Example System



A $\Rightarrow$ $M\frac{d^2\psi_a}{dt^2} = -K\psi_a + K(\psi_b - \psi_a)$

B $\Rightarrow$ $M\frac{d^2\psi_b}{dt^2} = -K\psi_b - K(\psi_b - \psi_a)$

Symmetric Analysis

Mode 1: $\psi_b = \psi_a$

$$\ddot{\psi}_a = \frac{-K}{M}\psi_a$$

$$= -\omega_1^2\psi_a$$

$$\omega_1^2 = \frac{K}{M}$$

$$B_1 = A_1$$

$(\psi_a)_1 = A_1\cos(\omega_1 t+\psi_1)$

$(\psi_b)_1 = A_1\cos(\omega_1 t+\psi_1)$

Mode 2: $\psi_b = -\psi_a$

$$\ddot{\psi}_b = \frac{K}{M}\psi_a + \frac{2K}{M}\psi_a$$

$$= \frac{3K}{M}\psi_a = -\omega^2\psi_a$$

$$\omega_2^2 = \frac{3K}{M}$$

$$B_2 = -A_2$$

$(\psi_a)_2 = A_2\cos(\omega_2 t+\psi_2)$

$(\psi_b)_2 = -A_2\cos(\omega_2 t+\psi_2)$